

HR Analytics – Employee Attrition Prediction & Decision Support

Domain: Human Resources Analytics & Machine Learning

1. Introduction

Employee attrition is an important HR challenge because frequent employee resignations can increase recruitment costs, training requirements, workload for existing staff, and loss of organizational knowledge. This project develops an HR analytics solution that studies historical employee data to identify patterns associated with attrition and estimate the risk of future attrition. The system combines exploratory data analysis, machine learning, explainable AI, and interactive business intelligence to convert employee data into understandable HR insights.

2. Abstract

The project, titled **Employee Attrition Prediction and HR Decision Support System**, analyzes employee attributes such as department, job role, income, overtime, satisfaction, tenure, and promotion history. The workflow begins with data preprocessing and exploratory analysis to understand attrition patterns. Classification models including Logistic Regression, Decision Tree, and Random Forest are trained to estimate employee attrition risk. Model performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. SHAP (SHapley Additive exPlanations) is incorporated to explain the factors influencing model predictions. Finally, Power BI is used to create an interactive dashboard for HR-oriented analysis and risk visualization. The resulting system is intended as a decision-support tool; predictions indicate risk based on historical patterns and do not guarantee that an employee will resign.

3. Tools Used

Tool / Technology	Purpose
Python	Data analysis and machine learning
Pandas, NumPy	Data loading, cleaning and transformation
Matplotlib, Seaborn	EDA and statistical visualization
Scikit-learn	Classification models and evaluation
SHAP	Explainable AI and prediction interpretation
Jupyter Notebook / VS Code	Development and experimentation
Power BI	Interactive HR analytics dashboard
Joblib	Saving the trained machine-learning model

4. Steps Involved in Building the Project

- **Dataset Selection:** An HR employee attrition dataset containing demographic, job, compensation, satisfaction, overtime, tenure, and other employee attributes was selected.
- **Data Preprocessing:** The dataset was inspected for missing values and duplicates. Non-informative identifier/constant columns were removed, categorical variables were encoded, and Attrition was converted into a binary target (1 = left, 0 = stayed).
- **Exploratory Data Analysis:** Attrition was analyzed across departments, job roles, overtime, income, age, job satisfaction, years at company, and promotion history using statistical summaries and visualizations.
- **Model Development:** Logistic Regression, Decision Tree, and Random Forest classifiers were trained using an 80:20 stratified train-test split.
- **Model Evaluation:** The models were compared using accuracy, precision, recall, F1-score and ROC-AUC. A confusion matrix and ROC curve were used to examine classification performance.
- **Explainable AI:** SHAP was applied to the Random Forest model to identify the features contributing to overall and individual attrition-risk predictions.
- **Power BI Dashboard:** A dashboard was prepared to visualize workforce metrics, attrition patterns, important HR factors, predicted risk levels, and employee-level prediction information.
- **Reporting:** Model results and analytical findings were documented along with practical, evidence-based attrition prevention suggestions and project limitations.

5. Conclusion

The completed project demonstrates how HR data can be transformed into an end-to-end analytics and decision-support solution. Exploratory analysis provides visibility into historical attrition patterns, machine learning provides a systematic method for estimating attrition risk, and SHAP makes the model more interpretable by showing which features contribute to predictions. Power BI further converts the analytical output into an accessible dashboard for HR users. The system should be used to support human decision-making rather than to make automatic employment decisions. Future improvements could include larger and more representative datasets, calibrated risk probabilities, model monitoring, and deployment through a secure HR application.